BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Paper II (Objective Type) — Ist - A - Exam 2023

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8191

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. A function f : x -> y defined by f(x) = a is called :

(A) Linear Function
(B) Constant Function
(C) Identity Function
(D) Implicit Function

Q2. Lim(x->inf) (x + e)/(x - e) = :

(A) 0
(B) infinity
(C) 1
(D) -1

Q3. The derivative of (x^2 - 4)/(x + 2) is equal to :

(A) -2
(B) 2
(C) 1
(D) 0

Q4. d/dx(e^x - e^-x) = :

(A) Sinhx
(B) Coshx
(C) 2 Sinhx
(D) 2 Coshx

Q5. sqrt(1 - x^2) d/dx(Cos^-1 x + Sin^-1 x) =

(A) 0
(B) 1
(C) 1/sqrt(1 - x^2)
(D) +2/sqrt(1 - x^2)

Q6. If f(x) = Sinx then f'(pi/2) =

(A) -1
(B) 0
(C) 1
(D) infinity

Q7. Integral dx/(x lnx) =

(A) lnx + C
(B) 1/x + C
(C) ln(lnx) + C
(D) (lnx)^2/2 + C

Q8. Integral from -pi to pi of Sinx dx = :

(A) 1
(B) -1
(C) 2
(D) 0

Q9. Integral e^-x(Cosx - Sinx) dx =

(A) e^-x Sinx + C
(B) e^-x Cosx + C
(C) -e^-x Sinx + C
(D) -e^-x Cosx + C

Q10. The solution of differential equation xdy + ydx = 0 :

(A) y = lnx + c
(B) y = ln(cx)
(C) y = ce^x
(D) xy = a

Q11. The equation of line x/(PSec alpha) + y/(PCosec alpha) = 1 is called :

(A) Symmetric Form
(B) Intercept Form
(C) Normal Form
(D) Slope Intercept Form

Q12. Slope of line with inclination 45 deg is :

(A) 1
(B) 0
(C) 1/sqrt(2)
(D) 1/sqrt(3)

Q13. Equation of Horizontal Line through (a, b) is :

(A) x = a
(B) x = b
(C) y = a
(D) y = b

Q14. (0, 0) is not a solution of the inequality :

(A) x - y < 1
(B) 2x + y < 1
(C) -2x + y + 1 > 0
(D) -2x + y < -1

Q15. Length of Tangent from (1, 1) to circle x^2 + y^2 - 2x + 3y + 6 = 0 :

(A) 1
(B) 2
(C) 3
(D) 4

Q16. Focus of the Parabola x^2 = -16y is :

(A) (0, 4)
(B) (0, -4)
(C) (4, 0)
(D) (-4, 0)

Q17. Co-Vertices of Ellipse x^2/b^2 + y^2/a^2 = 1; (a > b) are :

(A) (0, +-a)
(B) (+-a, 0)
(C) (0, +-b)
(D) (+-b, 0)

Q18. The vertices of Hyperbola x^2/4 - y^2/9 = 1 is :

(A) (0, +-3)
(B) (+-3, 0)
(C) (0, +-2)
(D) (+-2, 0)

Q19. |Cos alpha i + Sin alpha j + 0 K| = :

(A) 0
(B) 1
(C) 2
(D) -1

Q20. Length of Vector 2i - j - 2k is :

(A) 3
(B) -3
(C) sqrt(5)
(D) -sqrt(5)

Mathematics (Subjective) — Ist - A - Exam 2023

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: It is compulsory to attempt any (8 - 8) Parts each from Q.No. 2 and Q.No.3 while attempt any (9) Parts from Q.No.4. Attempt any (3) Questions from Part - II. Write same Question No. and its Part No. as given in the Question Paper — Part - I — 25 x 2 = 50

Q2.

Q.No.2

(i) Define Even Function and give one example.

(ii) Without finding Inverse, state Domain and Range of f^-1 where f(x) = 2 + sqrt(x - 1)

(iii) Indicate Solution Region by shading the Inequality 3x + 7y >= 21

(iv) Evaluate Lim(x->0) (1 - Cos2x)/x^2

(v) Find Derivative of sqrt(x) by definition.

(vi) Differentiate (x^2 + 1)^2/(x^2 - 1) w.r.t. x

(vii) Find dy/dx if x^2 - 4xy - 5y = 0

(viii) Differentiate Cos sqrt(x) + sqrt(Sinx) w.r.t. 'x'

(ix) If f(x) = ln(sqrt(e^2x + e^-2x)) then find f'(x).

(x) If y = Sinh^-1(x/2) then find dy/dx

(xi) If x = at^2, y = bt^4, then find y2

(xii) Define Feasible Region and Feasible Solution.

Q3.

Q.No.3

(i) Evaluate Integral (ax + b)/(ax^2 + 2bx + c) dx

(ii) Evaluate Integral sqrt(1 + Sinx) dx

(iii) Evaluate Integral e^x(Cosx - Sinx) dx

(iv) Evaluate Integral 1/(x lnx) dx

(v) Evaluate Integral from -1 to 3 of (x^3 + 3x^2) dx

(vi) Evaluate Integral from 0 to 3 of dx/(x^2 + 9)

(vii) Find the Integral Integral ax/sqrt(a^2 - x^4) dx

(viii) Find Direction Cosine of r = xi + yj + zk

(ix) Find a vector of length 5 in the direction opposite that of v = i - 2j + 3k

(x) If a + b + c = 0, then prove a x b = b x c = c x a

(xi) Find alpha so that alpha i + j, i + j + 3k and 2i + j - 2k are Coplaner.

(xii) Find the volume of Parallelepiped for u = i - 2j + 3k, v = 2i - j - k, w = j + k

Q4.

Q.No.4

(i) The Coordinates of a point P are (-2, 6). The axes are translated through the point O'(-3, 2). Find the Coordinates of Point P referred to new axes.

(ii) By means of slopes show that the points A(-4, 6), B(3, 8), C(10, 10) lie on the same line.

(iii) Find the equation of Horizontal Line through (7, -9)

(iv) Find the distance between the parallel lines 3x - 4y - 3 = 0, 3x - 4y + 7 = 0

(v) Whether point P(5, 8) lies above or below the line 2x - 3y + 6 = 0

(vi) Find the Area of a Triangular Region whose vertices are A(5, 3), B(-1, 2), C(4, 2)

(vii) Find the lines represented by 9x^2 + 24xy + 16y^2 = 0

(viii) Find the equation of a Circle centre at (5, -2) and Radius is 4

(ix) Find the equation of Normal to the Circle x^2 + y^2 = 25 at (5Cos theta, 5Sin theta)

(x) Find the length of Tangent drawn from (-5, 4) to the Circle 5x^2 + 5y^2 - 10x + 15y - 131 = 0

(xi) Find the Equation of Parabola with Focus (3, 2) and Directrix is x = -2

(xii) Find Foci and Vertices of an Ellipse x^2 + 4y^2 = 16

(xiii) Find the Equation of Hyperbola with Foci (+-5, 0), Vertex (3, 0)

> Note: Part - II — 3 x 10 = 30

Q5. (Marks 8) (a) Evaluate the Limit Lim(x->0) (e^(1/x) - 1)/(e^(1/x) + 1); x < 0 (5) (b) Differentiate w.r.t. x, Sec^-1((x^2 + 1)/(x^2 - 1)) (5)

Q6. (Marks 8) (a) Evaluate Integral 2x/(1 - Sinx) dx (5) (b) Find an Equation of the Perpendicular bisector of the Segment Joining the points A(3, 5) and B(9, 8) (5)

Q7. (Marks 8) (a) Find the area between the X-axis and the curve y = sqrt(2ax - x^2) When a > 0 (5) (b) Graph the feasible region of the following system of Linear Inequalities and find the Corner Points: 2x + 3y <= 18, x + 4y <= 12, 3x + y <= 12, x >= 0, y >= 0 (5)

Q8. (Marks 8) (a) If y = e^x Sinx, then show that d^2y/dx^2 - 2dy/dx + 2y = 0 (5) (b) Find an Equation of the Circle passing through A(3, -1), B(0, 1) and having Centre at 4x - 3y - 3 = 0 (5)

Q9. (Marks 8) (a) Find the Focus, Vertex and Directrix of Parabola. Sketch the graph x^2 - 4x - 8y + 4 = 0 (5) (b) By Vector Method, prove that in any Triangle b^2 = c^2 + a^2 - 2ca cosB (5)